



EcoTrack

VNR Design-A-Thon 2026

AI-Powered Carbon Intelligence for MSMEs

Simplifying Sustainability for Small Businesses
with **Blockchain Verification**.

Team Bottlers



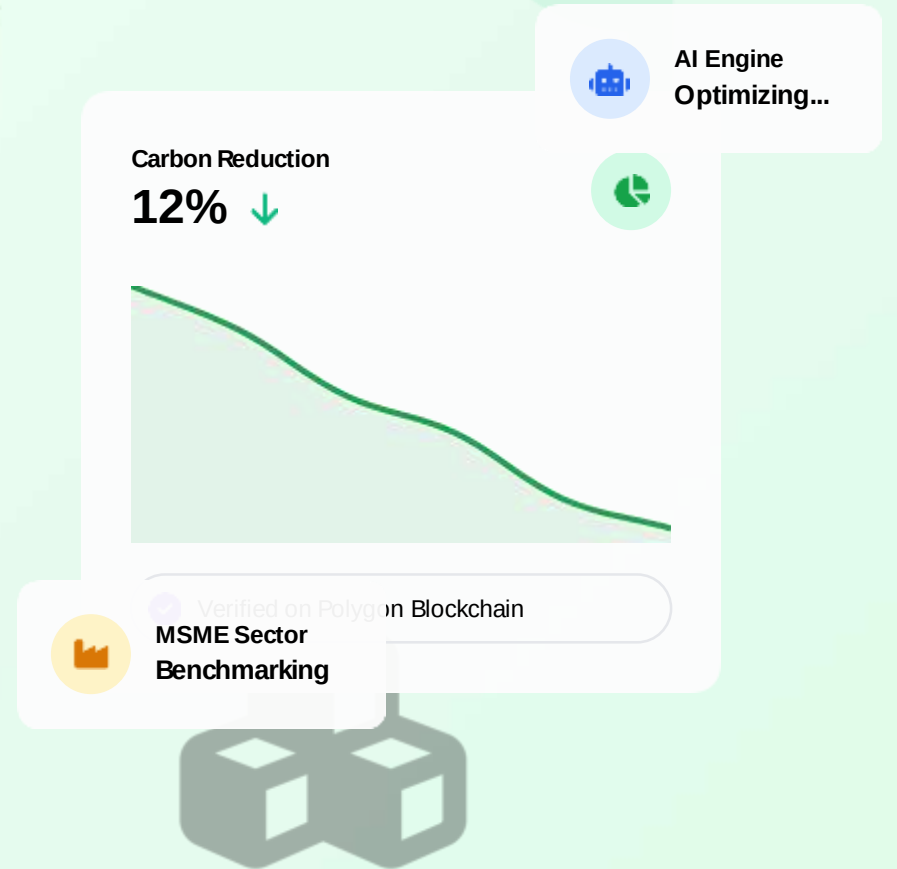
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Democratizing Carbon Intelligence for India's MSME Sector



Purpose

Enable India's 63–69 million MSMEs to measure, manage, and reduce carbon emissions affordably, addressing critical regulatory compliance and supply-chain pressures.



Methodology

Combines AI-driven analytics for automated Scope 1–3 calculations and hotspot detection with blockchain verification to ensure tamper-proof reporting and transparency.



Key Features

Automated data ingestion, MSME-specific industry benchmarking, and AI-powered reduction recommendations prioritized by cost and environmental impact.

Sector Impact Scope

~135 MtCO₂e



Total annual emissions from India's MSME sector in 2022 that EcoTrack aims to address through digital intervention.

EcoTrack Approach

₹500

Per Month

Affordable Access

100%

Verified

Blockchain Backed



Target Reach

63-69 Million MSMEs



The Backbone of India's Economy



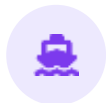
~69M

Total MSMEs



~30%

GDP Contribution



~46%

Export Share



~250M

Employment



Rising Regulatory Pressure: SEBI's BRSR framework mandates ESG disclosures for top listed entities, creating a cascade effect where large buyers demand carbon data from MSME suppliers.



Grid Emissions Context: Accurate accounting requires India-specific factors. The Central Electricity Authority (CEA) recommends a grid emission factor of **0.716 tCO₂/MWh** (FY 2022-23).

The Affordability Gap

Enterprise Tools

\$50k - \$250k/yr

MSME Budget

< \$1k/yr

Existing solutions are priced for Fortune 500 companies, not small businesses.

“ The Core Question

"How can a small manufacturing unit with 20 employees afford to track and reduce its carbon footprint when tools are designed for large enterprises?"



Problem Statement



The Challenge

Critical Issues & Pain Points



High Emission Impact

MSMEs emitted **~135 MtCO₂e** in 2022 and consume >25% of industrial energy.



Manual Calculation Errors

Spreadsheet-based tracking is error-prone, time-consuming, and lacks Scope 3 visibility.



No Benchmarking

Lack of MSME-scale industry data prevents setting realistic reduction targets.



Insight Gap

No actionable, costed insights to prioritize emission reduction investments.



Greenwashing Risks

Limited means to verify sustainability claims and reports authentically.



Target Users

Who Needs This Solution

Primary Segments

Manufacturing

Retail

Food & Beverage

Organization Size



5 - 500 Employees

Micro, Small & Medium Enterprises

Key Constraints



Limited sustainability budgets



No dedicated ESG teams

Market Size

63-69 Million

Potential businesses in India



Existing Solutions

Why They Fail MSMEs



Enterprise Platforms

✖ **Too Expensive**

Costing **\$50k+ per year**, these tools are built for complex corporate structures, not agile MSMEs.



Excel Calculators

✖ **Manual & Static**

Requires manual data entry, prone to human error, lacks automation, and provides zero AI-driven insights.



Generic Carbon APIs

✖ **Lack Localization**

Missing India-specific emission factors (CEA) and localized MSME benchmarking data.

"The market lacks an affordable, localized, and intelligent tool for Indian MSMEs."



Proposed Solution - EcoTrack

The EcoTrack Promise

Measure. Reduce. Verify.

A comprehensive carbon intelligence platform designed to make sustainability accessible for every Indian MSME.

Rs

Affordable SaaS

Starting at just **₹500/month**, removing the cost barrier for small businesses to access enterprise-grade tools.



AI Intelligence

Automated calculation for Scope 1, 2 & 3 with smart reduction recommendations prioritized by ROI.



Blockchain Verify

Tamper-proof reporting on **Polygon** ensuring authenticity and preventing greenwashing.



Why EcoTrack?



India-Specific Engine

Built with localized emission factors (CEA) and specific MSME sector benchmarks.



Actionable Savings

AI suggests cost-saving measures, not just carbon data, directly impacting the bottom line.



On-Chain Trust

First MSME-focused solution with blockchain verification to build supply chain trust.

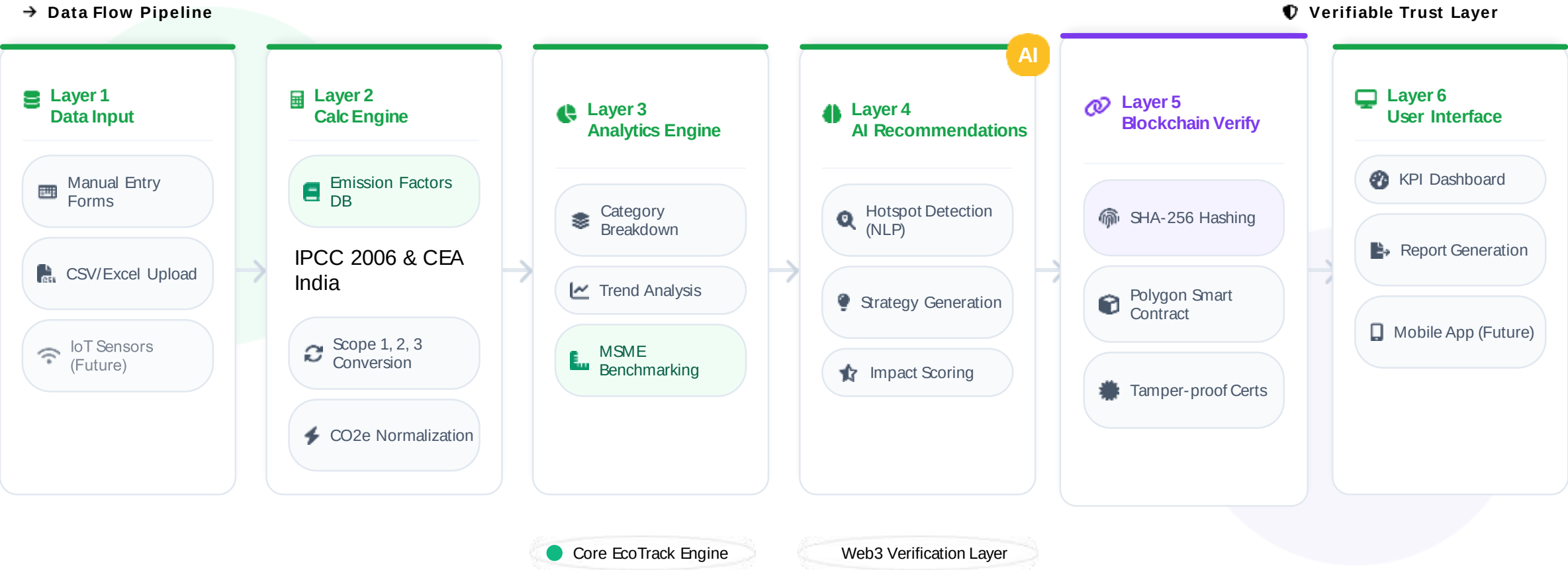


Mobile-First Design

Designed for business owners on the go, accessible anytime, anywhere.

Competitive Edge







UI / UX

Frontend Architecture

 Next.js 14 + TypeScript

 Tailwind CSS for Responsive UI

 Recharts for Visualizations

 Mobile-First Design System



Server

Backend Infrastructure

 Node.js with Express

 RESTful API Architecture

 PostgreSQL Database

 Supabase (Managed DB/Auth)



DevOps

Deployment & CI/CD

 Vercel (Frontend Hosting)

 Railway (Backend Services)

 Microservices Architecture



AI / ML Engine

Intelligent Insights

 Python (Scikit-learn, TensorFlow)

 K-Means Clustering for Benchmarking

 NLP for Strategy Generation



Blockchain

Trust Layer

 Polygon Mumbai Testnet

 Solidity Smart Contracts

 Ethers.js Web3 Integration



Key Features Deep Dive



Automated Carbon Calculation

Scope 1, 2, 3 Intelligence



Inputs: Monthly electricity, fuel, travel, waste data via manual entry or CSV upload.



Uses official CEA grid emission factor (0.716 tCO₂/MWh) for precise Indian context.

Real Calculation Example

1,000 kWh Electricity



716 kg CO₂e

Based on CEA Baseline Database v19.0



Industry Benchmarking

Comparative Analytics



Percentile ranking against peers by sector (e.g., Manufacturing) and size.



Identify "Best-in-Class" performers to set realistic reduction targets.

Peer Comparison

Your Performance

Top 25% of Sector



AI-Driven Recommendations

Smart Reduction Strategies



NLP engine detects hotspots and suggests 8-10 actionable reduction measures.



Prioritized by impact (High/Med/Low) and ROI payback period.

AI Suggestion



Switch to LED Lighting

Saves ~120 kg CO₂/month for typical small unit.



Blockchain Verification

Immutable Trust Layer



Every report generated creates a unique SHA-256 hash stored on Polygon.

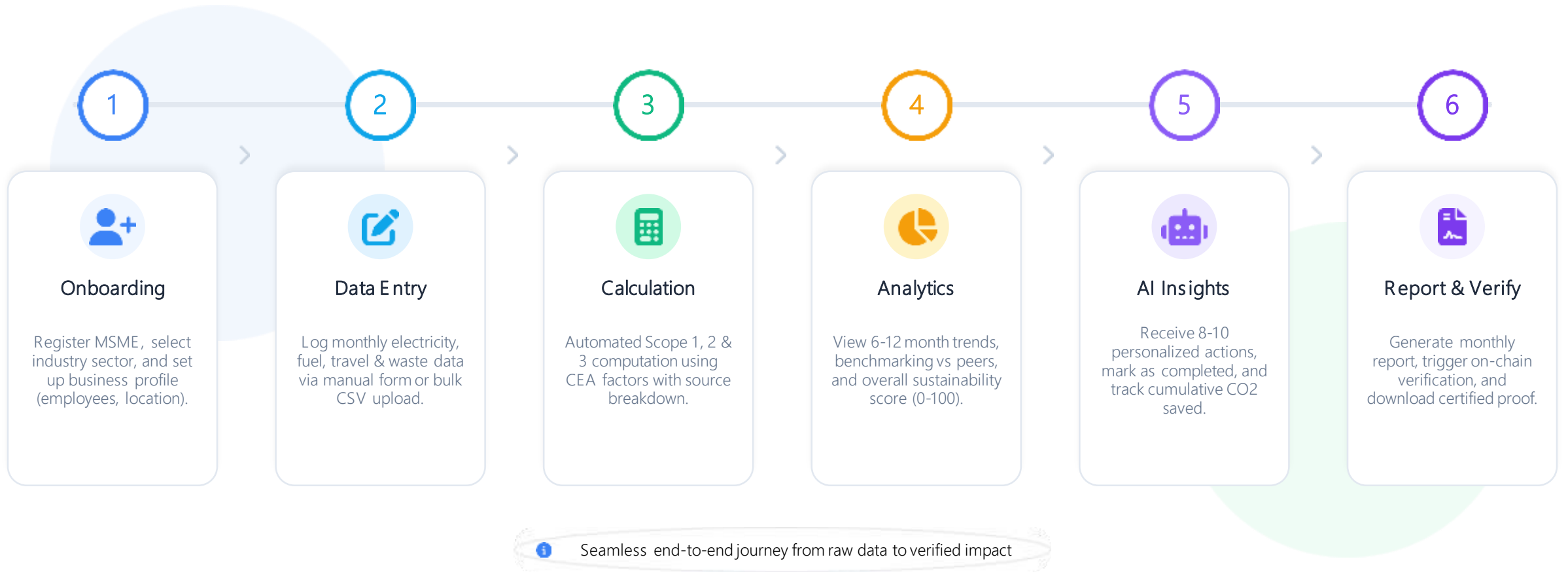


Provides tamper-proof audit trail and downloadable verified certificates.

On-Chain Proof

0x7f9...a2b4

Stored on Polygon Mumbai Testnet





Prototype Demo

1 Dashboard Overview

Live View

Total Emissions

4.2 tCO₂e

Sust. Score

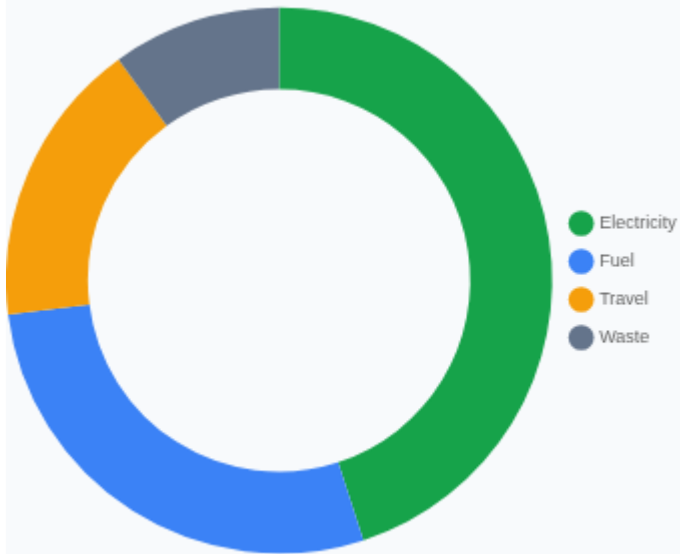
72/100

Benchmark

-18% vs Peers

Report Status

✓ Verified On-Chain



2 AI Recommendations

Smart Engine

Action Plan

8 Actions



Switch to LED lighting

HIGH

Replace 40 cfl units in production floor.

🌿 Save ~120 kg CO₂/mo



Optimize HVAC Schedule

MED

Reduce operation by 2 hours daily.

🌿 Save ~80 kg CO₂/mo



Install Solar Rooftop

HIGH

Transition to renewable energy.

🌿 Save ~200 kg CO₂/mo



Waste Segregation

LOW

Implement bin system for recycling.

🌿 Save ~15 kg CO₂/mo

3 Blockchain Verification

Polygon Net



Sustainability Report Verified
March 2026 • Manufacturing Sector

Transaction Hash (SHA-256)

0x71c9...3f2a8b9e1d4c5

Block Number

#24,891,003



Certificate



Immutable Record • Timestamped

Last updated: Just now

[View Details](#) →



Market Opportunity

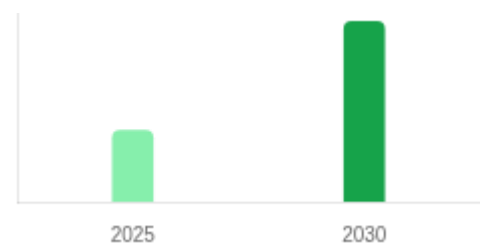
Immense Scale & Growth

69M+

MSMEs in India (Target Market)

Target: 3.15M

5% Adoption in 5 Years



Global Market ~\$38.14B by 2030



Monetization

SaaS Subscription Model

Freemium

₹0

Basic calculations, 1 user

Pro Tier

₹500/mo

AI Recommendations + Blockchain

Enterprise

₹2,000/mo

Multi-location + API Access

Strategic Revenue:



Bank partnerships (Green Loans)



White-label for Associations



Go-to-Market

Phased Rollout Strategy



Phase 1: Pilot

50 Mfg MSMEs in Pune/Mumbai cluster testing.



Phase 2: Partnerships

Collaborate with FISME, Laghu Udyog Bharati.



Phase 3: Scale

National rollout via digital channels & OEM partners.



Competitive Edge

Why EcoTrack Wins



Affordability

~10x cheaper than enterprise tools (\$50k+ vs ₹6k/yr).



Localization

India-specific emission factors & MSME benchmarking.



Trust Layer

First MSME solution with blockchain verification.

"Built specifically for the Bharat MSME ecosystem, not adapted from enterprise tools."

Environmental Impact

15-25%

Reduction in Year 1

For engaged MSMEs via efficiency & fuel switch

~5 Mt

Annual CO2e Saved

Scenario: 1M MSMEs × 20% of 25t baseline

Social Impact

Democratizing Sustainability: Making enterprise-grade carbon intelligence accessible to small businesses.

Green Finance Access: Verified reports enable MSMEs to qualify for lower-interest green loans.

Supply Chain Inclusion: Helps small vendors meet ESG requirements of large corporate buyers.

Regulatory Alignment

SEBI BRSR / BRSR Core

Ready for standardized disclosures required by listed entities and their value chains.

CBAM Readiness

Prepares exporters for Carbon Border Adjustment Mechanism compliance.

Supply Chain ESG

Supports Scope 3 reporting across complex MSME supply networks.

Scalability & Tech

Cloud-Native

API-First

Millions of Users

Integration Roadmap:

Tally

Zoho Books

Multilingual Support:

Hindi

Tamil

Telugu



Conclusion



EcoTrack bridges the affordability gap in carbon tracking for MSMEs by combining **AI-driven analytics** with **blockchain transparency**. We are purpose-built for the Indian ecosystem, utilizing localized emission factors and benchmarking to empower 63 million businesses.



Key Takeaways

- ✓ **Automated & Affordable:** Removes manual barriers with a SaaS model starting at just ₹500/month.
- ✓ **Actionable Intelligence:** AI suggests prioritized, cost-saving reduction strategies.
- ✓ **Verified Trust:** Industry-first blockchain verification tailored specifically for MSME compliance.
- ✓ **Scalable Growth:** Cloud-native architecture with a robust B2B partnership GTM strategy.



Future Scope



IoT Integration: Real-time energy monitoring sensors for precise data ingestion.



Scope 3 Expansion: Deep supply chain tracking for Tier-2/Tier-3 suppliers.



Carbon Trading: Tokenization of carbon credits on Polygon for marketplace trading.



Ecosystem Integration: Direct API plugins for Tally and Zoho Books accounting software.



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Dashboard

March 2026

Generate Report

TOTAL EMISSIONS

4.2 tonnes CO2e

12% less than last month

SUSTAINABILITY SCORE

72 / 100

Good standing

VS INDUSTRY AVERAGE

-18%

Below sector average

REPORT STATUS

Verified

On-Chain

Last verified: March 2026

Emission Sources



Electricity 45% Fuel 28% Travel 17% Waste 10%

Monthly Trend



NAVIGATION

Dashboard

Log Data

Analytics

AI Actions

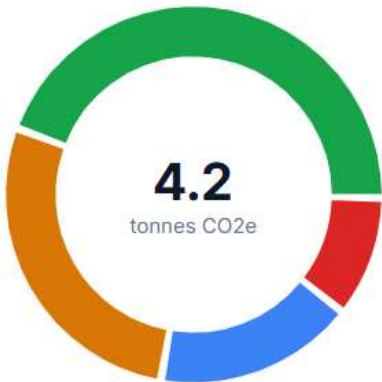
Reports

Blockchain

Goals

Settings

Emission Breakdown



Electricity	1890 kg		45%
Fuel	1176 kg		28%
Travel	714 kg		17%
Waste	420 kg		10%

12-Month Emissions



Scope Distribution



AI Actions

High

Saves 120 kg CO2/month

Replace fluorescent lighting with LED throughout the facility

LED bulbs consume up to 75% less energy than traditional fluorescent tubes and last significantly longer.

✓ Mark as Done

High

Saves 80 kg CO2/month

Reduce air conditioning operation by two hours daily

Adjust HVAC schedules to turn off 30 minutes before closing and delay startup in mornings.

✓ Mark as Done

Completed

Saves 200 kg CO2/month

Transition to a certified renewable energy provider

Switch your electricity contract to a 100% renewable energy tariff for immediate Scope 2 reduction.

✓ Completed

Medium

Saves 60 kg CO2/month

Optimize last-mile delivery routing using software

Implement route optimization to reduce fuel consumption and delivery time by up to 20%.

✓ Mark as Done

Completed

Saves 30 kg CO2/month

Migrate all documentation and billing to digital formats

Eliminate paper invoices, contracts, and internal memos by transitioning to cloud-based systems.

✓ Completed

Low

Saves 20 kg CO2/month

Install motion-activated lighting in low-traffic areas

Deploy occupancy sensors in storage rooms, hallways, and restrooms to reduce idle lighting.

✓ Mark as Done

Completed

Saves 15 kg CO2/month

Replace travel-based meetings with video conferencing

Low

Saves 10 kg CO2/month

Implement composting program for organic office waste

- NAVIGATION
- Dashboard

Log Data

Analytics

AI Actions

Reports

Blockchain

Goals

Settings

Log Activity Data

Record your business operations for March 2026

Electricity

Electricity consumed (kWh)

Enter kWh

Renewable energy percentage



0%

Fuel and Gas

Diesel (litres)

Enter litres

Petrol (litres)

Enter litres

LPG (kg)

Enter kg

Travel and Logistics

Distance travelled (km)

Enter km

Transport mode

Car

Waste generated (kg)

Enter kg



NAVIGATION

- Dashboard
- Log Data
- Analytics
- AI Actions
- Reports
- Blockchain
- Goals
- Settings

Verification Center

Carbon reports are cryptographically signed and immutably stored on the Polygon blockchain network

- 1

Report Finalized
Monthly data locked and hashed with SHA-256
- 2

Hash Stored
Transaction submitted to Polygon smart contract
- 3

Certificate Issued
Tamper-proof record with transaction ID

March 2026 — Verified

Polygon Mumbai Testnet

Report Hash	0x3a8f9c2d...e4b1d92c
Transaction ID	0x9b2e4f1a...c8d30e7b
Block Number	47,382,910
Timestamp	March 15, 2026 at 14:32 UTC

View on PolygonScan

Download Certificate

Verification History

PERIOD	EMISSIONS	HASH	STATUS	ACTION
March 2026	4.2t	0x3a8f9c2d...e4b1d92c	Verified	View
Feb 2026	4.0t	0x71d41c...562821	Verified	View